

NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA
DEPARTMENT OF ELECTRICAL ENGINEERING
MIDTERM EXAMINATION-I: Date: 20-02-2026
B.Tech. 2nd Semester

Subject: Electrical Circuit Theory
Time Allotted: 50 min.

Subject Code: EEPC 101
Max Marks: 15

Answer Any Three Questions

- Q1. Draw the Thevenin Equivalent of the circuit given in Fig. Q1 and find the load current I_R . (5)
- Q2. Determine the current through R_L in the circuit given in Fig. Q2 using Norton's Theorem. (5)
- Q3.(a) Write the Mesh current Equations in Matrix Form for the given Network shown in Fig. Q3.a (2.5)
- (b) For an electric network, a set of node voltage equations in matrix form is given below.
Construct the network taking into account the fact that between nodes 2 and 3 there is a practical voltage source of emf 32 V and internal resistance as derived from the equations given in Matrix Q3b. (2.5)
- Q4. (a) State & Prove Maximum Power Transfer Theorem. (4)
- (b) Define the Reciprocity Theorem. (1)

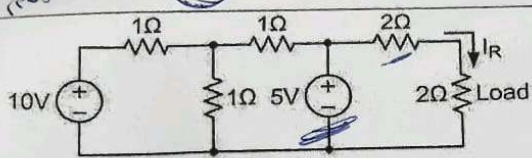


Fig. Q1

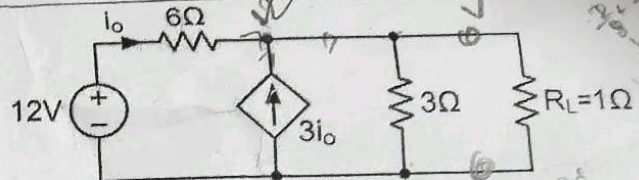


Fig. Q2

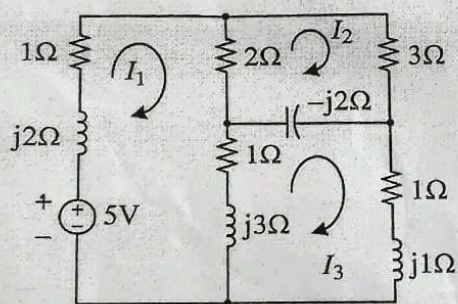


Fig. Q3. a

$$\begin{bmatrix} 7/8 & -1/4 & -1/8 \\ -1/4 & 29/80 & -1/16 \\ -1/8 & -1/16 & 3/16 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} -4 \\ 2 \\ 3 \end{bmatrix}$$

Matrix Q3.b

